

Experiment: 6

5G NR Frame Structure Analysis with Different Subcarrier Spacing

OBJECTIVE 1:

CODE:

```
clc;
```

```
clear;
```

```
close all;
```

```
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
```

```
mu = 0:4; % Numerology Index
```

```
bar(scs)
```

```
grid on
```

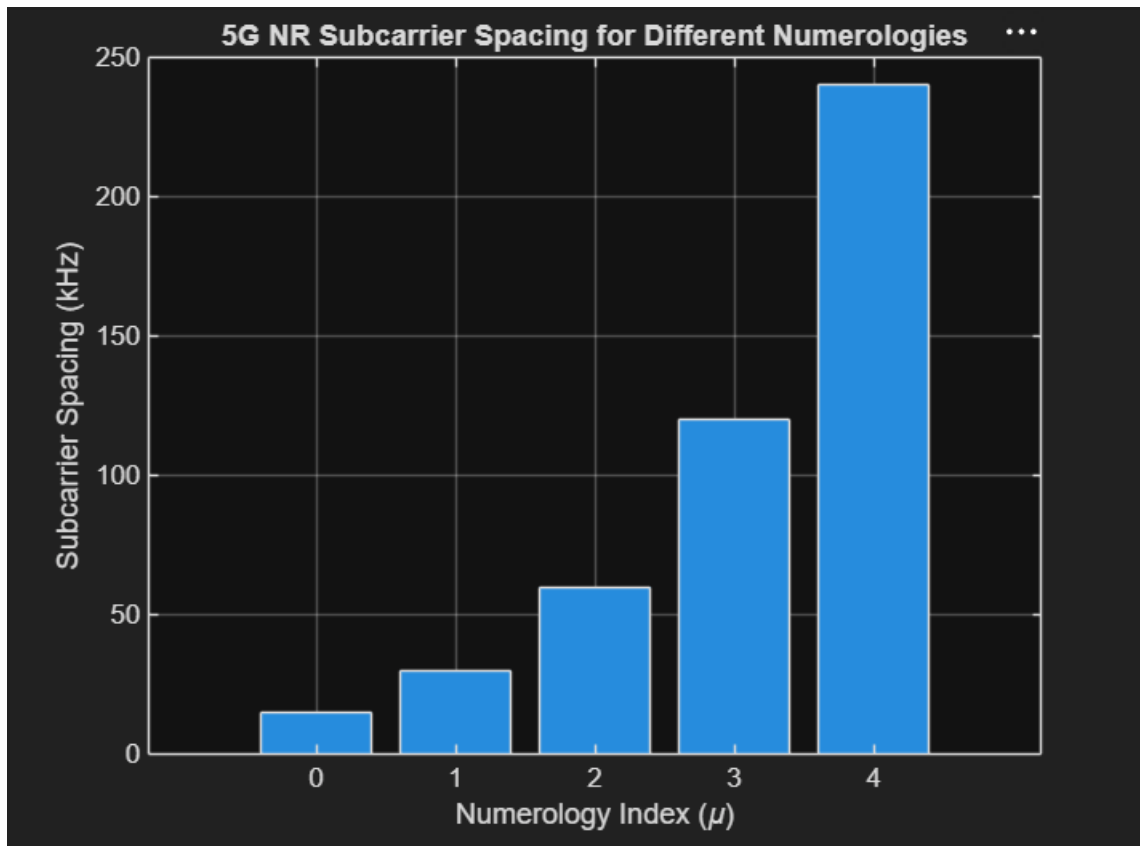
```
set(gca,'XTick',1:5)
```

```
set(gca,'XTickLabel',mu)
```

```
xlabel('Numerology Index ( $\mu$ )')
```

```
ylabel('Subcarrier Spacing (kHz)')
```

```
title('5G NR Subcarrier Spacing for Different Numerologies')
```



OBJECTIVE 2:

CODE:

```
clear;
```

```
close all;
```

```
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz) slot = [1  
0.5 0.25
```

```
0.125 0.0625]; % Slot Duration (ms)
```

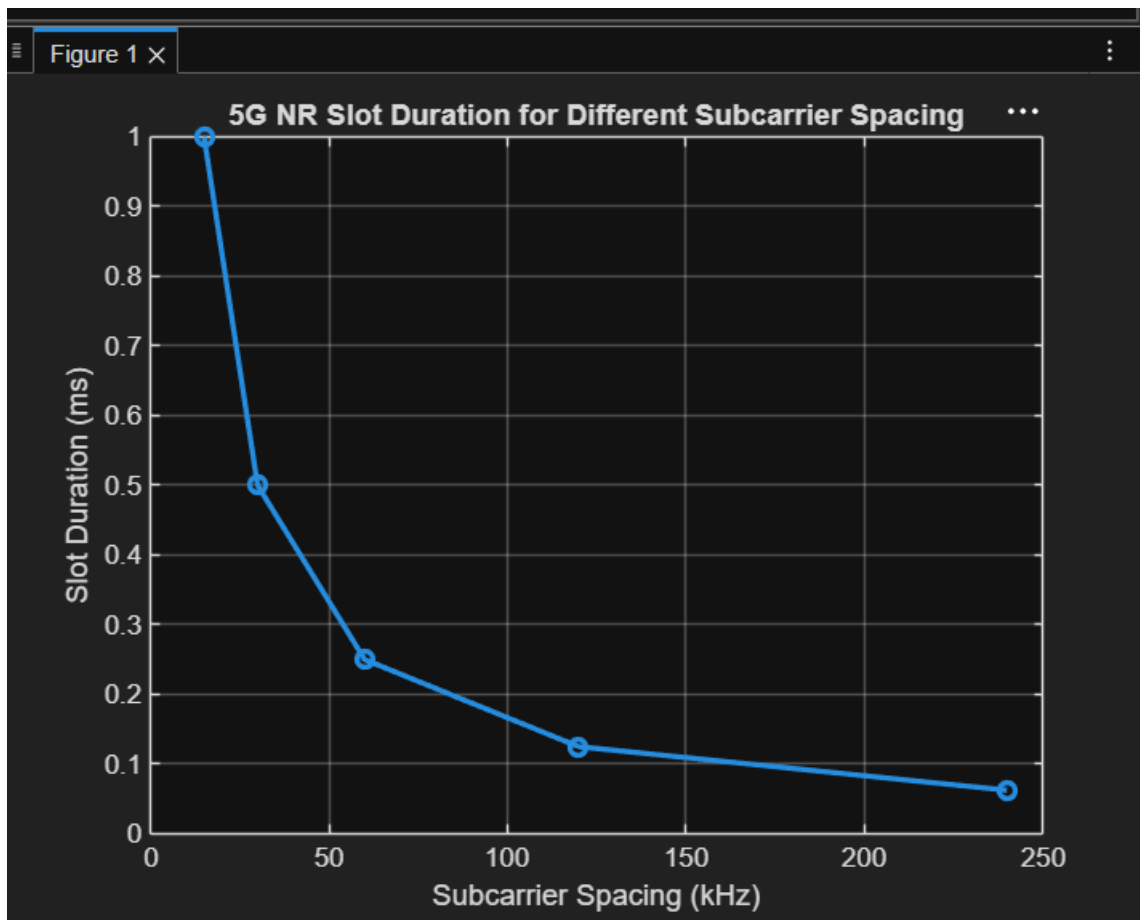
```
plot(scs, slot, 'o-', 'LineWidth', 2)
```

```
grid on
```

```
xlabel('Subcarrier Spacing (kHz)')
```

```
ylabel('Slot Duration (ms)')
```

title('5G NR Slot Duration for Different Subcarrier Spacing')



OBJECTIVE 3:

CODE:

```
close all;
```

```
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
```

```
symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
```

```
figure
```

```
bar(scs, symbols)
```

```
grid on
```

```
xlabel('Subcarrier Spacing (kHz)')
```

```
ylabel('Number of OFDM Symbols')
```

```
title('OFDM Symbols per Slot for Different Subcarrier  
Spacing')
```

```
ylim([0 16])
```

